Practical 1: Create a Simple SOAP Web Service

Objective: Create a SOAP web service in Java that converts **Dollar to Rupees**.

Step 1 — Open NetBeans

1. Start **NetBeans IDE**
2. Wait for the IDE to load completely.

Step 2 — Create a New Project

1. Click **File**
2. Click **New Project**

A window will open.

Now select:

Category: Java Web or Web

Project: Web Application

Click **Next**

Step 3 — Enter Project Details

Project Name: CurrencyService

Project Location: Leave default

Click **Next**

Step 4 — Select Server

Server: Apache Tomcat or GlassFish

Java Version: Java EE 7 Web or EE 8 Web

Click **Finish**

NetBeans will now create your project.

Step 5 — Create Web Service

In the **Projects panel (left side)**:

1. Right click the project - CurrencyService
2. Click: New → Web Service

Fill the form: Web Service Name -> Currency

Package Name: server or leave it to default

Click **Finish**

Note: If anything like this "Selected server doesn't seem to support JSR-109… Do you want to create METRO web services configuration file (sun-jaxws.xml)?**" come's select -> Click Yes**

What Happens Next

After clicking **Yes**:

1. NetBeans will create the **Web Service class**
2. You will see something like: Currency.java

inside your project.

Then you can continue with the next step:

Add the method:

```
public double DollarToRupee(double dollar) {
   return dollar * 80;
}
```

Step 6 — Add Web Service Operation

Now inside the generated file:

1. Right click **inside the code**
2. Click

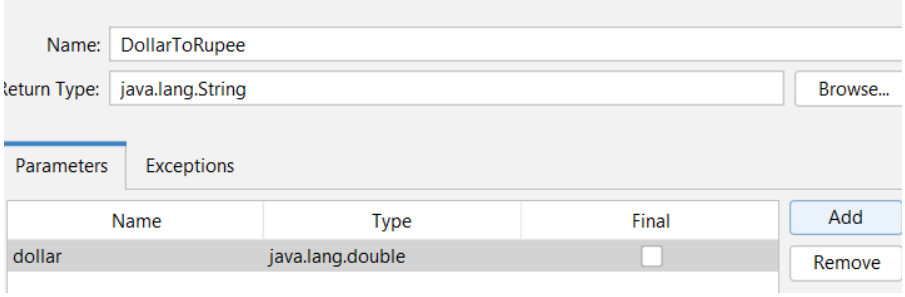
Insert Code

3. Select

Add Web Service Operation

Fill the form: Method Name-> DollarToRupee

Add parameter: Name: dollar, Type: double



Click **OK**

NetBeans will auto-generate code.

Step 7 — Write the Logic

Inside the method, replace the code with:

```
package server;
import javax.jws.WebService;
import javax.jws.WebMethod;
@WebService(serviceName = "Currency")
public class currency {
  @WebMethod(operationName = "DollarToRupee")
public double DollarToRupee(double dollar) {
  return dollar * 80;
}}
```

Note that: Kindly change the code if necessary

Step 8 — Deploy the Project

1. Right click project: CurrencyService
2. Click -> Deploy

Wait until deployment finishes.

Step 9 — Test Web Service

1. Expand the project
2. Expand **Web Services**
3. Right click -> Currency

4. Click

Test Web Service

Browser will open automatically.

Step 10 — Run the Method

In browser:

1. Enter value: 10 or anyhting
2. Click **DollarToRupee**

Output should be: 800

Expected Result

Dollar value converts to Rupees using the web service.

Example:

Input	Output
5	400
10	800

Practical 1 – Part 2

Create Java Client using JSP

Step 1 — Create Input Page

Inside NetBeans:

1. Expand your project
2. Expand **Web Pages**

Right click **Web Pages**

New → JSP

Name: input.jsp

Click **Finish**

Step 2 — Write Input Form

Replace code in **input.jsp** with:

```
<html>
<body>
<h2>Dollar to Rupee Converter</h2>
<form action="output.jsp">
Enter Dollar:
<input type="text" name="d">
<br><br>
<input type="submit" value="Convert">
</form>
</body>
</html>
```

Step 3 — Create Output Page

Again create new JSP.

New → JSP

Name: output

Click **Finish**

Step 4 — Create Web Service Client

Now we connect JSP to the web service.

Right click project: New → Web Service Client

Select **Browse**

Choose your service: CurrencyService

It will automatically detect: http://localhost:8080/CurrencyService/_currency?wsdl

Package Name: client

Click **Finish**

Now NetBeans will generate client classes automatically.

Step 5 — Drag Operation to JSP

Expand: Web Service References

Then expand: Currency

You will see the method: DollarToRupee

Now:

👉 Drag **DollarToRupee**
👉 Drop it inside **output.jsp body**

NetBeans will auto-generate code.

Step 6 — Modify the Generated Code

Change the parameter line to:

```
double result = port.dollarToRupee(Double.parseDouble(request.getParameter("d")));
```

Then print result:

```
out.println("Converted Rupees: " + result);
```

code:

```
<%@page contentType="text/html" pageEncoding="UTF-8"%>
<!DOCTYPE html>
<html>
<head>
<title>Output</title>
</head>
<body>
<h2>Conversion Result</h2>
<%
try {
   double dollar = Double.parseDouble(request.getParameter("d"));
   client.Currency_Service service = new client.Currency_Service();
   client.Currency port = service.getCurrencyPort();
   double result = port.dollarToRupee(dollar);
   out.println("Dollar: " + dollar + "<br>");
   out.println("Rupees: " + result);
} catch(Exception e) {
   out.println("Error: " + e);
}%>
```

```
</body>
</html>
```

Step 7 — Run Project

Right click: input.jsp

Click: Run File -> if running the file is not doing anything use below link kindly change the project name if u are not dumb.

Browser opens: http://localhost:8080/CurrencyService/input.jsp

Good 👍 Now we finish **Practical 1 – Python Client**.
This part will **consume the Java SOAP Web Service from Python**.

Your Java service is already running at:

http://localhost:8080/CurrencyService/_currency?wsdl

Practical 1 – Python Client

Step 1 — Install Python

Install Python if it is not installed.

Check installation: python --version

Step 2 — Install SOAP Library

Install **zeep** (Python SOAP client).

Open **Command Prompt** and run: pip install zeep

If pip fails: python -m pip install zeep

Step 3 — Create Python File

Create a file: client.py

You can create it in: Desktop / Documents / any folder

Step 4 — Write Python Code

Paste this code inside **client.py**.

```
from zeep import Client
wsdl_url = "http://localhost:8080/CurrencyService/_currency?wsdl"
client = Client(wsdl_url)
d = float(input("Enter Dollar: "))
result = client.service.dollarToRupee(d)
print("Rupees =", result)
```

Step 5 — Run the Python Program

Open terminal in the same folder.

Run: python client.py

Example: Enter Dollar: 10

Output: Rupees = 800

Important ⚠️

Before running Python:

✓ NetBeans project must be **running**
✓ Web service must be **deployed**
✓ URL must open in browser

Test URL:

http://localhost:8080/CurrencyService/_currency?wsdl

If this opens → Python will work.

Expected Output

Example run:

Enter Dollar: 5

Rupees = 400

Practical 1 – .NET Client

Step 1 — Open Visual Studio

1. Start **Visual Studio**

2. Click **Create a new project**

Step 2 — Create Console Application

In the search box type: console

Select: Console App (.NET Framework)

Click **Next**

Step 3 — Enter Project Details

Project Name: CurrencyClient

Click: Create

Visual Studio will create the project.

Step 4 — Add Web Service Reference

In **Solution Explorer (right side)**

1. Right click the project
2. Click

Add → Service Reference

A new window opens.

In the **Address box** paste your WSDL URL:

http://localhost:8080/CurrencyService/_currency?wsdl

Click: Go

Visual Studio will detect the service.

Namespace: CurrencyRef

Click:OK

Now a folder **Service References** will appear in your project.

Step 5 — Write C# Code

Open: Program.cs

Replace the code with:

```
using System;
using CurrencyClient.CurrencyRef;
namespace CurrencyClient
{
   class Program
   {
      static void Main(string[] args)
      {
         CurrencyClient client = new CurrencyClient();
         Console.Write("Enter Dollar: ");
         double d = Convert.ToDouble(Console.ReadLine());
         double result = client.dollarToRupee(d);
         Console.WriteLine("Rupees = " + result);

         Console.ReadLine();
      }   }}
```

Step 6 — Run the Program

Click **Start** or press:

F5

Example run:

```
Enter Dollar: 10
Rupees = 800
```